Network Protocol

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1. Encoding and Decoding

Every network protocol message sent in the network begins with a four capital letter sequence which matches one of the predefined network commands, presented later in this document. Depending on the command itself, it can be followed by any number of arguments. In case it is necessary an argument can be encoded by simply preceding one digit in front of it, which relays the length of the argument together with this one encoded digit. As long as the amount of arguments is set, this can be expanded to more than one argument. An abstract representation of a network message:

<4 letter protocol><digit><1. argument><digit><2. argument>...<digit><n. argument><msg>

A received network protocol message can be decoded as long as the amount of arguments is known.

2. Network Protocol Commands

LGIN - Login

This command signals that a new player / client has connected to the server. No data string has to be attached to the command.

LOGT - Logout

This command signals that a player wants to logout from the server and leave the game. The server closes the connection to the client and updates the client list. No data string has to be attached.

QUIT - Quitting the game

This command transmits that a player wants to leave the game. The server makes sure to close the connection and update the player list. No data string has to be attached.

NAME - Set Username

This command signals the player's intention to change their username. The server checks if the username is already being used by some other client and changes it if needed. The attached data string contains the desired new username. This command is only sent from clients to the server.

HSCR - Highscores

This command shows the player's intention to have a look at the high score list. The server transmits the high score list to the client. If the command is sent from the client to the server, then no data string is required.

CELY - Create new a Lobby

This command signals that the client wants to create a new lobby. The server creates the new lobby and adds the client to it. This command is sent from the client to the server and no data string is needed.

JOLY - Join Lobby

This command indicates that a player wants to join an already existing lobby. The name of the lobby is attached in the data string. This command is only sent from client side to server side.

QILY - Leave Lobby

This command signals the player's intention to leave his lobby. The server displays a leaving message and removes the client from the lobby and the player list. This command is only sent from clients to the server.

NEWG - New Game

This command signals the player's intention to start a new game. The server now initiates a new game with the client's lobby.

BGIN - Begin

This command initializes a new instance of the game with a board and figures. This command can only be sent by the server. The data string contains all the game details.

ADLY - Add Lobby

This command signals that a new lobby has been created. The server informs the client. The new lobby is added to the lobby list. This command is only sent from the server side to the client side.

MOVE - Move

This command sends the movement of a chess figure. The data is formatted as follows:
"Starting Position" + "Ending Position" and encoded by the encoder.

SELT - Select

This message signals that a player has selected a specific figure on the board. This command is sent from the client to the server.

SLTD - Selected

This command tells all clients which figure has been selected by the current player. This command is sent from the server to the clients.

ILMV - Illegal Move

This command alerts the player when his desired move is not possible (illegal).

LMOV - Last Moves

This command sends a detailed report of the moves, made in the last turn, to be displayed in the player moves window. The data string is encoded by the encoder.

TURN - Turn

This command tells all the players that a new turn has begun and who's turn it is. The current player's ID is transmitted as the data.

STAT - Status

This command relays the current status of a lobby chess board.

GOVR - Game Over

This command signals that the game is over. The command is sent from the server to the clients and the attached string contains the information about the game result.

CHAT - Global Chat

This command signals that a client wants to send a message to all the connected clients. The server then transmits this message to all the clients that are in the global chat.

WCHT - Whisper Chat

This command signals that a client wants to send a private message to another client. This chat message is sent from a client to a server and then redirected to the recipient.

LCHT - Lobby Chat

This command signals that a client has sent a message in the lobby chat. The server sends the message only to the players who are connected to the same lobby.

ELOB - Empty Lobby

This command signals that a lobby is empty and needs to be removed from the lobby list. After receiving this command, the client removes the lobby from the local lobby list.

SLOB - Show Lobbies

This command shows all available lobbies to the player. The server sends a list of all lobbies as a data string encoded by the encoder.

SNAM - Show Name

This command is sent from the client to the server. The server then returns the client's username.

SLNA - Show Lobby Name

This command sends the name of the lobby the client currently is in, to the client as a data string. This command is sent from the client to the server.

SPLY - Show players

This command shows the client how many players are already in the lobby. It transmits a simple integer as data.

SONO - Sound

This command is used to tell the client when and which sound files to play. It is sent from the server side based on the current game stance. The data part of the message contains a string that tells the client which sound to play, for example:
"Start"
"Piece destroyed"
"Check"
"Checkmate"